TECHNICAL DILIGENCE REVIEW - RECORD 023

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-023

1. Scope of the component and licence review

The report at TDR-2026-31 was prepared by Pinehurst Systems Assurance for Ardent Peak and covers the third-party component inventory, licence exposure and contribution provenance of the codebase at Helioscope Robotics, Inc. as at 2026-07-01. The instruction behind the present report was narrow: examine the third-party component inventory, licence exposure and contribution provenance of the codebase and report by 2026-07-01. TDR-2026-31 recommends that 2 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. Deal implications in this report are modest in aggregate and specific in detail: 2 items for the closing checklist, 7 for the plan. S. Alderton has kept the working papers behind this report in a form that a second reviewer could re-perform.

The population for this report was the service inventory as at 2026-07-01; 17 entries were examined and the rest were reviewed only at inventory level. 6 sites contributed evidence to TDR-2026-31, and one further site was in acceptance testing and was excluded. The reviewer notes that the report at TDR-2026-31 was prepared while three remediation programmes were in flight, so several positions will have moved by completion. S. Alderton could not verify one class of claim in report TDR-2026-31, namely the behaviour of components deployed only at the two sites not visited. Pinehurst Systems Assurance raised each finding with the owning engineer before it was written up, and the responses are recorded.

In this report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. The present report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. The report at TDR-2026-31 keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. Where management has declined to act, the present report preserves the decision and the stated rationale so a post-close plan can revisit it. The evidence behind TDR-2026-31 is retained by Pinehurst Systems Assurance and is available to Ardent Peak on request.

This report carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The principal limitation on this report is time: 3 weeks was enough to test the control path and not enough to test the reporting path. Not every finding in this report is accepted by management: 2 are declined, and the declinatures are reproduced rather than argued away. One declinature in report TDR-2026-31 rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. The third-party component inventory, licence exposure and contribution provenance of the codebase is reported here at the level of detail Ardent Peak asked for and no deeper.

S. Alderton has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. In the report at TDR-2026-31 the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. S. Alderton interviewed 11 engineers and managers for the present report; two declined to be quoted on one topic and that is noted where it matters. Interviewees for TDR-2026-31 were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. The reviewer's tone in report TDR-2026-31 is deliberately flat; a finding is not made more serious by the words around it.

The report at TDR-2026-31 draws on 11 interviews, including the engineering owners of every component examined in detail. 11 interviews inform report TDR-2026-31. Where an interview is the sole source for a claim, the claim is graded limited. Two positions in the report at TDR-2026-31 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. 2 items in the report at TDR-2026-31 sit in a declined state, and the reviewer regards one of them as wrongly declined. The company's engineering leadership saw every finding in the report at TDR-2026-31 before delivery and disputed two of them.

Schedule T - Third-party component and licence inventory (TDR-2026-31)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically	approved 2023-	Costmap Fusion

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
				linked by the fusion pipeline	08-04	
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding across edge and cloud	approved 2024-04-08	Config Distribution
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip detector, not shipped	not approved	Wheel Odometry Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

2. Component inventory and licence classification

The non-licence risk on sqlite-embedded is a single-maintainer dependency with no succession plan. For sqlite-embedded the obligation is nothing beyond retaining the licence text with the source, and the reviewer tested whether it is met. Sqlite-embedded appears at OSS-010 in version 3.45.1; the usage recorded is on-robot local state. sqlite-embedded was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Tracing grid-render to its source gives public registry; the commercial licence was purchased in 2024. Grid-render also carries a single-maintainer dependency with no succession plan, which is a supportability question rather than a licence one. The approval position on grid-render is approved with conditions 2023-07-19. grid-render appears twice in the working papers for TDR-2026-31 because the first answer was corrected after a follow-up request.

The supply route for lidar-vendor-sdk is supplied by the sensor vendor. To publish the modified files it distributes is what lidar-vendor-sdk requires in practice. Lidar-vendor-sdk carries the approval entry approved 2023-05-22; the reviewer looked for a corresponding decision record. lidar-vendor-sdk appears twice in the working papers for TDR-2026-31 because the first answer was corrected after a follow-up request.

Where yaml-parse came from matters here: public registry, signature verified. The approval position on yaml-parse is approved 2023-09-29. Compliance with the yaml-parse terms requires nothing beyond retaining the licence text with the source. yaml-parse was tested rather than accepted, and the test is described in the working papers behind TDR-2026-31.

K8s-operator-framework shows an approval status of approved 2024-09-24 against the company's own policy. OSS-028 covers k8s-operator-framework at version 1.31.0, used cloud deployment controllers. On provenance, k8s-operator-framework shows public registry, signature verified. k8s-operator-framework illustrates why TDR-2026-31 separates the severity of a finding from the confidence behind it.

Usage of otel-instrument 1.24.0 is tracing across cloud services, and the inventory reference is OSS-023. Linking mode matters for otel-instrument, which is used tracing across cloud services. Otel-instrument obliges the company to hold a current commercial licence for each developer seat. otel-instrument is reported at the evidence strength available and has not been promoted to fit the narrative.

Wms-adapter-sdk version 2.2.0 is catalogued at OSS-017 and is used one customer's warehouse system adapter. Wms-adapter-sdk sits inside WMS Connector and is therefore in the shipped product rather than in tooling. Wms-adapter-sdk is licensed under commercial, per-seat, with copyleft exposure recorded as no. wms-adapter-sdk is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The licence on geom2d-fusion is LGPL-2.1; copyleft exposure is weak. The approval position on geom2d-fusion is approved 2023-08-04. Provenance of geom2d-fusion is public registry; upstream branch unmaintained since 2024. geom2d-fusion was tested rather than accepted, and the test is described in the working papers behind TDR-2026-31.

At OSS-008 the component is rtos-shim, pinned at 1.0.9 and used real-time helper layer on the robot. In Motion Planner the component rtos-shim is used real-time helper layer on the robot, which is the fact the licence analysis rests on. Meeting the terms on rtos-shim means to attribute the component in the customer-facing notice file. rtos-shim is one of the entries where TDR-2026-31 relies on the company's own measurement rather than an independent one.

Protobuf-runtime also carries a fork that must be maintained locally for as long as it is used, which is a supportability question rather than a licence one. Under BSD-3-Clause, protobuf-runtime is treated as no copyleft in the inventory. Approved 2024-04-08 is the approval entry for protobuf-runtime. protobuf-runtime was checked against the company's own register, which had not been updated since the change landed.

Schedule C - Producer and consumer interface contracts examined (TDR-2026-31)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	1 in the reviewed period
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	1 in the reviewed period
Costmap Fusion	motion planner	published schema	v3.9	not automated	1 in the reviewed period
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	2 in the reviewed period
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	2 in the reviewed period
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the pipeline	none
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	none
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	2 in the reviewed period
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	none
Charge Scheduler	fleet health monitor	request-response interface	v1.8	not automated	none
Safety Controller	drive power contactors	published schema	v2.5	advisory only	1 in the reviewed period
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	none
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	2 in the reviewed period
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	2 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	1 in the reviewed period
Lidar Driver	localization engine	published schema	v3.1	advisory only	2 in the reviewed period
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	2 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	2 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	none
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	2 in the reviewed period
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	2 in the reviewed period
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	1 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	none
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	2 in the reviewed period
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	none
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	none
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	1 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	none
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	none
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	none
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	2 in the reviewed period
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	1 in the reviewed period
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	none
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	none
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	none
Battery Management Service	charge scheduler	request-response interface	v4.11	not automated	2 in the reviewed period

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	1 in the reviewed period
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	1 in the reviewed period
Gripper Controller	safety controller fault input	published schema	v3.5	enforced on the producer side	2 in the reviewed period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	1 in the reviewed period
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	2 in the reviewed period
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	2 in the reviewed period
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	none
Event Bus	every consuming service	published schema	v1.5	advisory only	none
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	2 in the reviewed period
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	none
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	2 in the reviewed period
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	2 in the reviewed period
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	none
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	1 in the reviewed period
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	1 in the reviewed period
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	none
Alarm Router	paging provider	streamed topic	v2.3	not automated	2 in the reviewed period
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	2 in the reviewed period
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	2 in the reviewed period
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	1 in the reviewed period
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	2 in the reviewed period
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	none
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	1 in the reviewed period
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	none
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	2 in the reviewed period
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	none
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	none
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	none
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	2 in the reviewed period

3. Copyleft exposure and the shipped-software boundary

The non-licence risk on safety-controller-toolchain is no identified risk beyond ordinary version currency. Safety-controller-toolchain is licensed under commercial, development use only, with copyleft exposure recorded as no. Meeting the terms on safety-controller-toolchain means nothing beyond retaining the licence text with the source. safety-controller-toolchain was reviewed against the position twelve months earlier, and the direction of travel is favourable.

OSS-027 records pager-provider-sdk 3.4.0 against Alarm Router, alarm dispatch. Supplied by the paging provider is the provenance recorded for pager-provider-sdk. The non-licence risk on pager-provider-sdk is an upstream project

with no active maintainer. pager-provider-sdk is included because a reader of TDR-2026-31 should be able to disagree with the inference without repeating the work.

The inventory entry OSS-020 is cell-model-lib 1.8.0, battery state estimation. Apache-2.0 governs cell-model-lib, which the inventory marks as no copyleft. Licence text and the actual usage of cell-model-lib were read together, and attribution present. cell-model-lib would be a candidate for the first post-close engineering review under any reasonable plan.

The non-licence risk on wheel-slip-model is an upstream project that is active but changes interface frequently. Wheel-slip-model is recorded as not approved in the component approval log. Version 0.3.1 of wheel-slip-model is the catalogued build at OSS-025, used prototype slip detector, not shipped. wheel-slip-model was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Provenance of kafka-client is public registry, signature verified. Because kafka-client is used event bus producer and consumer clients, its licence terms reach the delivered software. Apache-2.0 governs kafka-client, which the inventory marks as no copyleft. kafka-client was cross-checked against the engineering record supplied for TDR-2026-31, and the two agree.

The distribution question for pointcloud-utils turns on the fact that it is used point cloud filtering in fusion. File-level copyleft; three modified files must be published and the reviewer could not evidence publication is the reviewer's observation on pointcloud-utils. Policy compliance for pointcloud-utils rests on its approval entry, which is approved 2024-03-14. pointcloud-utils is not material on its own and belongs in this report because of what it reveals about the surrounding process.

OSS-003 covers scanmatch-legacy at version 1.2.0, used linked into the localization fallback path. Under GPL-2.0, scanmatch-legacy is treated as strong copyleft in the inventory. Where scanmatch-legacy came from matters here: vendored source copied from a project archive in 2022. scanmatch-legacy would move from a concern to a verified defect if one further measurement were supplied.

In the approval log, grpc-transport is approved 2024-05-30. Against grpc-transport the reviewer would add that no notice obligations beyond attribution. To share alike any derived asset it distributes is what grpc-transport requires in practice. grpc-transport is reported at the evidence strength available and has not been promoted to fit the narrative.

One point about chartkit is not resolved: no security updates are available upstream; a replacement is on the debt register. Meeting the terms on chartkit means to publish the modified files it distributes. Chartkit also carries an upstream project that is active but changes interface frequently, which is a supportability question rather than a licence one. chartkit was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Jwt-verify was assessed for support as well as licence risk, and it carries a build-time dependency whose absence would stop the pipeline rather than the product. Provenance of jwt-verify is public registry, signature verified. Linking mode matters for jwt-verify, which is used token verification in the identity service. jwt-verify was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2026-31)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	second delivery path
SEC-2025-02	High	142	30 days	no	none recorded
SEC-2025-03	Medium	87	90 days	yes	manual review before release
SEC-2025-04	Medium	43	90 days	yes	none recorded
SEC-2025-05	High	76	30 days	no	second delivery path
SEC-2025-06	Medium	22	90 days	yes	restricted role membership
SEC-2025-07	Medium	24	90 days	yes	audit trail only
SEC-2025-08	Medium	154	90 days	no	restricted role membership
SEC-2025-09	Medium	308	90 days	no	second delivery path
SEC-2025-10	Low	167	180 days	yes	network segmentation
SEC-2025-11	Low	78	180 days	yes	monitoring alert on use

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-12	Low	354	180 days	no	restricted role membership
SEC-2026-01	High	232	30 days	no	audit trail only
SEC-2026-02	High	127	30 days	no	audit trail only
SEC-2026-03	High	27	30 days	yes	audit trail only
SEC-2026-04	Medium	135	90 days	no	second delivery path
SEC-2026-05	Medium	100	90 days	no	restricted role membership
SEC-2026-06	Medium	192	90 days	no	none recorded
SEC-2026-07	Medium	158	90 days	no	second delivery path
SEC-2026-08	Medium	151	90 days	no	audit trail only
SEC-2026-09	Low	150	180 days	yes	network segmentation
SEC-2026-10	Low	78	180 days	yes	restricted role membership
SEC-2026-11	Low	200	180 days	no	second delivery path
SEC-2026-12	Low	252	180 days	no	monitoring alert on use
SEC-2026-13	Low	24	180 days	yes	manual review before release
SEC-2026-14	Low	128	180 days	yes	restricted role membership
SEC-2026-15	Low	127	180 days	no	network segmentation
SEC-2026-16	Medium	148	90 days	no	manual review before release
SEC-2026-17	High	147	30 days	no	network segmentation
SEC-2026-18	Low	305	180 days	no	audit trail only
SEC-2026-19	Medium	116	90 days	no	second delivery path
SEC-2026-20	Low	97	180 days	no	network segmentation

4. Where third-party code sits inside the product

Robot Shadow Store shows 68 per cent line coverage in the pipeline, and the figure is respectable but excludes the fallback path, which is where the known defect sits. Behind Robot Shadow Store sits Redis 7 as the hot path with PostgreSQL 14 as the durable record; the service itself is Go 1.22 on cloud, containers with a sharded cache. Underlying Robot Shadow Store in this report are the shadow consistency note, staleness measurements and a rebalance observation. Robot Shadow Store carries a single measurable objective, namely that shadow state is no more than two seconds behind the robot at p95. The ageing position for Robot Shadow Store is none identified, and the reviewer checked it against the vendor statements on file. Robot Shadow Store is treated as material to the investment case rather than as a routine engineering item.

9 items sit in the Robot Shadow Store backlog; the entry that bears on the investment case is that a shard rebalance briefly serves stale shadow state without marking it stale. Robot Shadow Store ships weekly; the reviewer sampled the last twelve releases. Robot Shadow Store owns its own data and exposes it only through a versioned interface, which the reviewer read. Deployment of Robot Shadow Store is cloud, regional, sharded by site, which sets the blast radius of any defect in it. Taking the evidence together, Robot Shadow Store the component would not block the transaction, and it would appear on any competent post-close engineering plan. Robot Shadow Store was compared with the equivalent position at two other components, and it is neither the best nor the worst.

WMS Connector was weighed on its coverage, its defect backlog and its dependency position, and the design is sound and the test evidence is thinner than the design deserves. WMS Connector does not share a schema with any consumer; every read by the order broker and the billing usage meter goes through the published interface. WMS Connector carries 23 open defects, among them a case where duplicate order suppression relies on a customer-supplied identifier that two tenants reuse. WMS Connector has a single named owner, Tomas Villareal of Integrations, which is not true of every service reviewed. The cadence recorded for WMS Connector is every two weeks, per tenant on request, which matches the observed release history. WMS Connector was checked against the corresponding entry in the company's own register, and the register was current.

Supportability of WMS Connector turns on one fact: one customer integration still runs on a connector build from 2024. The manifests place WMS Connector cloud, one deployment per customer tenant. Java 21 is the implementation language for WMS Connector, which is hosted on cloud, containers on a managed Kubernetes cluster with PostgreSQL 14 plus a message outbox table behind it. Automated test coverage on WMS Connector stands at 69 per cent line coverage. Inputs to WMS Connector come from the customer warehouse management systems and the identity service, and its consumers are the order broker and the billing usage meter. WMS Connector carries a follow-up request that was satisfied after the draft of TDR-2026-31 was circulated.

Telemetry Ingest commits that telemetry visible in the operator console inside 30 s of emission at p95, and the objective is measured continuously rather than sampled. For Telemetry Ingest the reviewer worked from the telemetry schema registry, retention configuration and a forced-offline buffer test. Telemetry Ingest's dependency ageing was checked: the stream processor runs a release that leaves community support in late 2026. Changes to Telemetry Ingest reach production weekly, with schema changes gated on a compatibility check. 19 defects remain open against Telemetry Ingest; the one the reviewer pressed on is that the edge buffer drops the oldest samples after four hours offline without raising an alert. Telemetry Ingest rests on a single source, and the reviewer has said so rather than implying corroboration.

One coupling concern applies to Telemetry Ingest: a time-series store with ninety-day retention and object storage beyond that is shared with a second service, so a schema change is not a local decision. Telemetry Ingest was assessed against the objective and the defect record, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. 61 per cent line coverage is reported for Telemetry Ingest, and the reviewer re-ran the measurement on a clean checkout. Telemetry Ingest runs cloud, one ingest endpoint per region with an edge buffer on the robot; that placement is what determines whether a fault is one robot or one region. Telemetry Ingest depends on the edge gateway agent and the event bus and is depended on by the fleet health monitor, the analytics warehouse loader and the alarm router. Telemetry Ingest illustrates why TDR-2026-31 separates the severity of a finding from the confidence behind it.

The on-robot camera buffer and the identity service feed Video Snippet Service; a failure in Video Snippet Service is felt by the remote operations console and the incident review workflow. Supportability of Video Snippet Service turns on one fact: a transcoder build with known but low-severity upstream advisories. The open defect backlog for Video Snippet Service holds 12 items, the most consequential being that the retention lifecycle was applied per bucket, so one region kept clips for ninety days. Video Snippet Service owns its own data and exposes it only through a versioned interface, which the reviewer read. The manifests place Video Snippet Service cloud, regional, with an on-robot capture buffer. Video Snippet Service was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Go 1.22 with a media pipeline is the implementation language for Video Snippet Service, which is hosted on cloud, containers plus a media transcoder with object storage with a thirty-day lifecycle and a PostgreSQL 14 index behind it. Video Snippet Service ships monthly; the reviewer sampled the last twelve releases. For Video Snippet Service the stated target is that a requested snippet is available to the operator inside 45 s, expressed against production traffic rather than a benchmark. Video Snippet Service was assessed against the objective and the defect record, and the reviewer accepts the engineering position while noting that the verifying test does not exist. Video Snippet Service sits at 43 per cent line coverage; the trend across the reviewed period was up by seven points. Video Snippet Service would move from a concern to a verified defect if one further measurement were supplied.

Billing Usage Meter has a single named owner, Nadia Semenova of Data and Telemetry, which is not true of every service reviewed. The ageing position for Billing Usage Meter is none identified, and the reviewer checked it against the vendor statements on file. Billing Usage Meter rests on the metering specification, three monthly reconciliations and the ledger immutability check so far as this review is concerned. The cadence recorded for Billing Usage Meter is monthly, aligned to the billing cycle, which matches the observed release history. Billing Usage Meter reports 79 per cent line coverage, measured by the same instrumentation the pipeline uses. Billing Usage Meter was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Deployment of Billing Usage Meter is cloud, regional, which sets the blast radius of any defect in it. Billing Usage Meter publishes a stable interface, though two fields used by the finance export and the customer usage report are documented only in code comments. The open defect backlog for Billing Usage Meter holds 7 items, the most

consequential being that a re-run of a failed metering window can double-count if the operator skips the reset step. Service-level intent for Billing Usage Meter states that metered usage reconciles to the order journal within 0.5 per cent each month. Billing Usage Meter is written in Go 1.22, executes on cloud, containers on a managed Kubernetes cluster, and keeps its state in PostgreSQL 14 with an immutable usage ledger. Billing Usage Meter was checked against the corresponding entry in the company's own register, and the register was current.

The ageing position for Lidar Driver is the vendor's software development kit is two major versions behind current, and the reviewer checked it against the vendor statements on file. Yuki Harada of Robot Firmware maintains Lidar Driver, and the service inventory entry was current at the review date. Lidar Driver shows 44 per cent line coverage in the pipeline, and coverage alone overstates the assurance here, because the bench suite is the real safety net. Lidar Driver owns its own data and exposes it only through a versioned interface, which the reviewer read. Underlying Lidar Driver in this report are the driver source, the vendor integration guide and a captured reconnect trace. Lidar Driver is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Lidar Driver carries a single measurable objective, namely that no more than one dropped scan per ten thousand at nominal spin rate. The topology for Lidar Driver is edge, one process per sensor, and the reviewer confirmed it against the deployment manifests. C++17 is the implementation language for Lidar Driver, which is hosted on on-robot Linux, user-space driver with no store; frames are handed to fusion in memory behind it. The costmap fusion pipeline and the localization engine rely on Lidar Driver, which in turn relies on the sensor hardware and the time synchronisation daemon. Lidar Driver releases as needed, historically twice a year. Lidar Driver would be a candidate for the first post-close engineering review under any reasonable plan.

Release Orchestrator sits at 55 per cent line coverage; the trend across the reviewed period was up by seven points. Release Orchestrator was checked for shared-database coupling, and the ownership boundary held. In production Release Orchestrator is cloud, single global deployment, unchanged since the last architecture review. Release Orchestrator sits between the build system, the simulation harness gate and the identity service and the OTA updater and the cloud deployment controllers, and that position is what makes its availability objective load-bearing. Hosting for Release Orchestrator is cloud, a pipeline runner with approval hooks, and its Go 1.22 code writes through to PostgreSQL 14 for release records and object storage for artefacts. Release Orchestrator was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

What the reviewer had for Release Orchestrator was the release policy, ninety release records and an audit of emergency-path usage; the conclusions below do not reach past it. Release Orchestrator has a single named owner, Colin Achebe of Developer Productivity, which is not true of every service reviewed. Developer Productivity tracks 16 open defects on Release Orchestrator, including the acknowledged case where an emergency release path allows a single approver, and it was used more often than the policy expects. Service-level intent for Release Orchestrator states that no artefact reaches production without a recorded approval and a passing gate set. For Release Orchestrator the supportability position is none identified. Release Orchestrator was checked against the corresponding entry in the company's own register, and the register was current.

Interface discipline around Charge Scheduler is adequate; the schema is versioned and change is gated on a compatibility check. Charge Scheduler was examined for unsupported components; none; the dependency set was refreshed in the 2026 platform upgrade. Automated test coverage on Charge Scheduler stands at 63 per cent line coverage. Fleet Services holds Charge Scheduler, and the ownership entry naming Nadia Semenova matches the on-call rotation. On paper Charge Scheduler releases fortnightly, without a formal approval gate until March 2026, and the observed practice did not diverge from that. Charge Scheduler was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Charge Scheduler sits between the battery management service and the site shift calendar and the fleet allocator and the fleet health monitor, and that position is what makes its availability objective load-bearing. Charge Scheduler runs cloud, regional, with a conservative local fallback policy on the robot; that placement is what determines whether a fault is one robot or one region. 7 defects remain open against Charge Scheduler; the one the reviewer pressed on is that opportunity-charge windows are not released when a shift ends earlier than planned. For Charge Scheduler the reviewer worked from the scheduler policy document and four sites' charge utilisation exports. The Charge Scheduler codebase is Python 3.11 throughout, deployed onto cloud, containers on a managed Kubernetes cluster, and its durable record is

PostgreSQL 14, shared instance with the allocator. Charge Scheduler is included because a reader of TDR-2026-31 should be able to disagree with the inference without repeating the work.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-31)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-200 tote mover	safety scanner	2.10.5	14	both rigs	one revision behind
HR-300 pallet mover	navigation lidar	3.10.7	12	both rigs	one revision behind
HR-200 tote mover	load cell deck	2.1.2	11	neither rig	current
HR-410 manipulator	charge contact assembly	1.6.1	7	rig one only	one revision behind
HR-320 pallet mover	load cell deck	2.13.2	8	neither rig	current
HR-200 tote mover	network radio	2.11.8	9	rig one only	current
HR-200 tote mover	depth camera	1.9.8	6	neither rig	current
HR-300 pallet mover	wheel encoder set	3.3.5	3	both rigs	current
HR-200 tote mover	depth camera	3.2.4	4	rig two only	current
HR-410 manipulator	load cell deck	1.3.5	5	both rigs	pre-production
HR-200 tote mover	depth camera	2.8.3	5	rig one only	current
HR-410 manipulator	load cell deck	3.8.0	13	neither rig	superseded
HR-240 tote mover	wheel encoder set	3.17.2	13	rig two only	pre-production
HR-200 tote mover	network radio	3.1.2	4	rig two only	superseded
HR-300 pallet mover	safety scanner	3.17.6	11	neither rig	pre-production
HR-200 tote mover	depth camera	3.14.0	6	rig one only	pre-production
HR-410 manipulator	network radio	2.3.1	10	rig two only	one revision behind
HR-200 tote mover	wheel encoder set	2.13.5	14	neither rig	current
HR-410 manipulator	compute module	3.1.5	4	neither rig	current
HR-410 manipulator	network radio	2.9.1	4	rig two only	one revision behind
HR-410 manipulator	network radio	3.14.4	9	rig one only	pre-production
HR-300 pallet mover	load cell deck	4.0.3	14	neither rig	one revision behind

4. Where third-party code sits inside the product - part 2 of 2

Where the 4. Where third-party code sits inside the product was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. The sequence is unbroken either side of it.

Definitions given once for the 4. Where third-party code sits inside the product are relied on again rather than restated, the compilation basis being identical. Pagination alone determines where it falls.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-31) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-320 pallet mover	network radio	2.0.2	12	rig two only	pre-production
HR-300 pallet mover	wheel encoder set	4.15.6	5	both rigs	pre-production
HR-300 pallet mover	charge contact assembly	1.5.6	6	neither rig	current
HR-300 pallet mover	inertial measurement unit	1.6.2	7	rig two only	pre-production
HR-320 pallet mover	safety scanner	1.7.2	7	rig two only	pre-production
HR-240 tote mover	navigation lidar	4.6.2	13	rig one only	current
HR-240 tote mover	charge contact assembly	3.5.8	11	rig two only	one revision behind
HR-320 pallet mover	depth camera	4.6.4	7	rig two only	one revision behind
HR-410 manipulator	wheel encoder set	3.0.2	12	both rigs	pre-production
HR-200 tote mover	inertial measurement unit	4.8.6	6	both rigs	one revision behind
HR-200 tote mover	E-stop harness	4.17.2	10	rig two only	pre-production
HR-300 pallet mover	network radio	3.1.7	3	neither rig	superseded
HR-300 pallet mover	wheel encoder set	3.18.0	10	rig one only	current
HR-300 pallet mover	navigation lidar	1.12.0	4	neither rig	pre-production
HR-320 pallet mover	load cell deck	3.6.0	13	neither rig	superseded
HR-320 pallet mover	network radio	3.0.1	10	rig one only	current
HR-240 tote mover	wheel encoder set	4.5.0	6	both rigs	one revision behind
HR-240 tote mover	safety scanner	4.15.2	12	rig two only	pre-production
HR-240 tote mover	inertial measurement unit	1.0.2	11	neither rig	current
HR-410 manipulator	compute module	4.5.4	10	neither rig	current
HR-300 pallet mover	charge contact assembly	3.17.8	4	rig one only	current
HR-320 pallet mover	compute module	2.12.6	2	rig two only	current

5. Provenance, approval and contributor assignment

Csv-export obliges the company to reproduce the notice text in the delivered documentation. Licence text and the actual usage of csv-export were read together, and attribution present. Csv-export was assessed for support as well as licence risk, and it carries an upstream project that is active but changes interface frequently. csv-export is the clearest instance in TDR-2026-31 of engineering practice being better than the documentation suggests.

The terms applying to image-signing are Apache-2.0; the copyleft classification is no. Approval for image-signing reads approved 2025-01-15, and the reviewer checked the approving record. Image-signing carries a practical obligation: to attribute the component in the customer-facing notice file. image-signing has no single accountable owner, and that is the finding the reviewer would press hardest.

Opencv-subset presents a fork that must be maintained locally for as long as it is used to the engineering team. To attribute the component in the customer-facing notice file is what opencv-subset requires in practice. Version 4.9.0 of opencv-subset is the catalogued build at OSS-011, used image preprocessing for grasp proposals. opencv-subset illustrates why TDR-2026-31 separates the severity of a finding from the confidence behind it.

Simulation Harness is where physics-core-fork is consumed, which determines whether distribution occurs at all. Physics-core-fork obliges the company to share alike any derived asset it distributes. Physics-core-fork was assessed for support as well as licence risk, and it carries an upstream project that is active but changes interface frequently. physics-core-fork is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Http-router reaches the customer only through Order Broker, and the reviewer confirmed there is no second path. OSS-022 covers http-router at version 1.9.4, used cloud service routing. Compliance with the http-router terms requires to exclude a particular build configuration from the shipped artefact. http-router is the clearest instance in TDR-2026-31 of engineering practice being better than the documentation suggests.

The open question on tsdb-client is that attribution present in the cloud notice file. The operational obligation attaching to tsdb-client is to keep the dynamic-linking boundary intact. The supply route for tsdb-client is public registry, signature verified. tsdb-client is the clearest instance in TDR-2026-31 of a control that exists on paper and is weaker in practice.

Warehouse-sim-assets was obtained as follows: downloaded from a public asset library. On warehouse-sim-assets the reviewer records that share-alike terms apply to derived assets; four internally modified assets are not published. Warehouse-sim-assets also carries no identified risk beyond ordinary version currency, which is a supportability question rather than a licence one. warehouse-sim-assets sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Beyond licensing, nav-graph-solver carries a fork that must be maintained locally for as long as it is used. The terms applying to nav-graph-solver are Apache-2.0; the copyleft classification is no. Tracing nav-graph-solver to its source gives public registry, signature verified. nav-graph-solver is the reason one section of TDR-2026-31 concludes narrowly rather than generally.

To share alike any derived asset it distributes is what pgbouncer-managed requires in practice. Attribution present is the reviewer's observation on pgbouncer-managed. Fleet Allocator is where pgbouncer-managed is consumed, which determines whether distribution occurs at all. pgbouncer-managed was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

On media-transcode the reviewer records that the build must exclude the GPL-configured components and the reviewer verified the current build flags. Beyond licensing, media-transcode carries a component with open advisories that do not affect the deployed configuration. In the approval log, media-transcode is approved with conditions 2024-08-16. media-transcode was inspected in the repository rather than described from the ticket, which is why the account is specific.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2026-31)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-23-01	the objective is met on the company's own	limited	is reflected in the post-close plan	a witnessed test

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
	measurement only			
OQ-23-02	two teams each believe the other owns this	strong	does not change the conclusion	one further interview
OQ-23-03	the declined item has no expiry date	limited	is reflected in the post-close plan	a re-run of the original test
OQ-23-04	the declined item has no expiry date	strong	would change the conclusion if resolved adversely	a witnessed test
OQ-23-05	two teams each believe the other owns this	limited	narrows the conclusion to the tested scale	one further interview
OQ-23-06	the compensating control was not tested	limited	would change the conclusion if resolved adversely	an independent measurement
OQ-23-07	the declined item has no expiry date	strong	is reflected in the post-close plan	a re-run of the original test
OQ-23-08	the measurement basis changed during the reviewed period	limited	does not change the conclusion	a document the company has not produced
OQ-23-09	the declined item has no expiry date	strong	does not change the conclusion	a document the company has not produced
OQ-23-10	the measurement basis changed during the reviewed period	strong	does not change the conclusion	a re-run of the original test
OQ-23-11	the supplier certificate was read rather than the test witnessed	moderate	would change the conclusion if resolved adversely	a decision from management
OQ-23-12	the compensating control was not tested	limited	is reflected in the post-close plan	a decision from management
OQ-23-13	the second bench rig does not cover this path	limited	is reflected in the post-close plan	one further interview
OQ-23-14	the declined item has no expiry date	limited	would change the conclusion if resolved adversely	a decision from management
OQ-23-15	the objective is met on the company's own measurement only	limited	does not change the conclusion	a re-run of the original test
OQ-23-16	the objective is met on the company's own measurement only	strong	is reflected in the post-close plan	a decision from management
OQ-23-17	the supplier certificate was read rather than the test witnessed	strong	leaves one section unconcluded	a re-run of the original test
OQ-23-18	the objective is met on the company's own measurement only	limited	does not change the conclusion	an independent measurement
OQ-23-19	the remediation landed without a regression test	moderate	narrows the conclusion to the tested scale	a document the company has not produced
OQ-23-20	the retention window closed before the raw data was requested	moderate	is reflected in the post-close plan	an independent measurement
OQ-23-21	the objective is met on the company's own measurement only	limited	leaves one section unconcluded	one further interview
OQ-23-22	the supplier certificate was read rather than the test witnessed	moderate	narrows the conclusion to the tested scale	a document the company has not produced
OQ-23-23	the measurement basis changed during the reviewed period	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-23-24	the remediation landed without a regression test	limited	would change the conclusion if resolved adversely	one further interview
OQ-23-25	the declined item has no expiry date	limited	would change the conclusion if resolved adversely	a decision from management
OQ-23-26	the retention window closed before the raw data was requested	moderate	does not change the conclusion	one further interview
OQ-23-27	the declined item has no	limited	would change the	a document the company

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
	expiry date		conclusion if resolved adversely	has not produced
OQ-23-28	the second bench rig does not cover this path	moderate	would change the conclusion if resolved adversely	a document the company has not produced
OQ-23-29	the declined item has no expiry date	limited	does not change the conclusion	an independent measurement
OQ-23-30	two teams each believe the other owns this	strong	leaves one section unconcluded	a document the company has not produced
OQ-23-31	two teams each believe the other owns this	moderate	is reflected in the post-close plan	a re-run of the original test
OQ-23-32	the objective is met on the company's own measurement only	limited	is reflected in the post-close plan	a re-run of the original test

6. Findings arising from dependency and secret scanning

On assessment finding SEC-2026-14 the conclusion is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. The practical consequence in finding SEC-2026-14 is that a manual corridor override cannot be attributed after the fact. Assessment finding SEC-2026-14 was opened on 2026-04-21 and states that traffic manager admin action lacked an audit record. 2026-06-18 is the closure position on Low-severity finding SEC-2026-14, which the ledger records as remediated. Low-severity finding SEC-2026-14 was checked against the corresponding entry in the company's own register, and the register was current.

Medium-severity finding SEC-2025-06 is not fully retired by its remediation: low; the index is scoped per tenant and the identifiers are opaque. Assessment finding SEC-2025-06 rests on the observation that enumeration confirmed; no clip content was retrievable; the reviewer holds that to be moderate. SEC-2025-06 against Video Snippet Service translates into the following exposure: a customer operator could enumerate the existence of clips in another tenant. SEC-2025-06 against Video Snippet Service was graded Medium, band 6.5, by Redwater Security engagement RW-2025-3. finding SEC-2025-06 was inspected in the repository rather than described from the ticket, which is why the account is specific.

At 5.2 the score on assessment finding SEC-2026-05 places it in the Medium group. Medium-severity finding SEC-2026-05 is assessed as follows: the fix is verified and the class of defect is not, since no test now prevents its return. Residual risk after remediation of SEC-2026-05 against Alarm Router is medium; per-region credentials are scheduled but not issued. What finding SEC-2026-05 enables is this: one leaked credential affects paging for every region. finding SEC-2026-05 would be a candidate for the first post-close engineering review under any reasonable plan.

Strong evidence supports assessment finding SEC-2025-11, namely that attribute absence confirmed; no working exploit chain was demonstrated. Reading SEC-2025-11, the defect is that console session cookie lacked a strict same-site attribute; the discovery date is 2025-09-25. Assessment finding SEC-2025-11 is assessed as follows: the severity is right and the ownership is unclear, since two teams each believe the other holds it. 2025-10-30 is the closure position on Remote Operations Console finding SEC-2025-11, which the ledger records as remediated. SEC-2025-11 was excluded from an earlier draft of TDR-2026-31 and reinstated after the reviewer obtained the supporting record.

Medium-severity finding SEC-2026-04 arrives as a Medium finding with a 5.6 score, and the reviewer did not adjust either. Read alongside the remediation record, SEC-2026-04 shows that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Remediated is the status of Medium-severity finding SEC-2026-04, dated 2026-05-20. The practical consequence in Medium-severity finding SEC-2026-04 is that customer commercial data is retained in operational logs. WMS Connector finding SEC-2026-04 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The remediation ledger shows assessment finding SEC-2026-12 as partially remediated, pending. Finding SEC-2026-12 matters because a disposed compute module carries a usable fleet credential. 2026-04-20 is the discovery date of SEC-2026-12 against Edge Gateway Agent, which finds that edge gateway agent retained mutual authentication keys after decommissioning. The evidence for SEC-2026-12 is moderate: keys recovered from a returned module. assessment finding SEC-2026-12 is treated as material to the investment case rather than as a routine engineering item.

At 3.2 the score on assessment finding SEC-2026-20 places it in the Low group. Assessment finding SEC-2026-20 sits against Payload Scale Interface and records that payload scale calibration record can be edited without an audit entry. The reviewer declines to soften SEC-2026-20: the remediation is adequate and the time to remediate is not. As at the review date SEC-2026-20 against Payload Scale Interface was accepted and the closure entry reads not planned. SEC-2026-20 would move from a concern to a verified defect if one further measurement were supplied.

Low-severity finding SEC-2025-10 was graded Low, band 3.7, by Redwater Security engagement RW-2025-3. 2025-09-24 is the discovery date of Low-severity finding SEC-2025-10, which finds that charge scheduler exposed an unauthenticated internal health endpoint with site names. Remediated is the status of finding SEC-2025-10, dated 2025-10-09. What remains after SEC-2025-10 against Charge Scheduler is remediated is low. SEC-2025-10 against Charge Scheduler is one of the entries where TDR-2026-31 relies on the company's own measurement rather than an independent one.

In SEC-2026-02 against Robot Shadow Store the exploitation path means any cluster workload could read fleet position data for all tenants. On Robot Shadow Store finding SEC-2026-02 the conclusion is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. For SEC-2026-02 against Robot Shadow Store the record shows that the tester read another tenant's shadow state from a test workload, which is moderate evidence and is graded separately from severity. Residual risk after remediation of Robot Shadow Store finding SEC-2026-02 is low; authorization is checked on both the cache and durable paths. SEC-2026-02 was checked against the company's own register, which had not been updated since the change landed.

On SEC-2026-06 against Simulation Harness the conclusion is that the item is open past its committed date and no revised date has been agreed. A compromised scenario could read production artefact storage is the exposure finding SEC-2026-06 describes. SEC-2026-06 is rated Medium with a numeric band of 6.1 on the tester's scoring scale. The remediation ledger shows Medium-severity finding SEC-2026-06 as partially remediated, pending. assessment finding SEC-2026-06 illustrates why TDR-2026-31 separates the severity of a finding from the confidence behind it.

What remains after SEC-2026-09 is remediated is low. At 3.9 the score on SEC-2026-09 against Fleet Health Monitor places it in the Low group. Fleet Health Monitor finding SEC-2026-09 stands remediated; 2026-05-08 is what the record gives for closure. A single query can exhaust the service's memory budget is the exposure Fleet Health Monitor finding SEC-2026-09 describes. Low-severity finding SEC-2026-09 is the reason one section of TDR-2026-31 concludes narrowly rather than generally.

2025-11-14 is the closure position on assessment finding SEC-2025-01, which the ledger records as remediated. For finding SEC-2025-01 the record shows that reproduced by the tester and confirmed in the token audit, which is strong evidence and is graded separately from severity. Service tokens are issued with a twelve-hour lifetime is the substance of assessment finding SEC-2025-01, raised on 2025-09-16. In finding SEC-2025-01 the exploitation path means a captured token remains usable long after the session that produced it. Identity and Authorization finding SEC-2025-01 has no single accountable owner, and that is the finding the reviewer would press hardest.

Residual exposure on SEC-2026-10 reads low; object lock is applied and verified nightly. The reviewer declines to soften SEC-2026-10 against Billing Usage Meter: the risk acceptance lacks an expiry date, which is a process finding in its own right. Billing Usage Meter finding SEC-2026-10 is rated Low with a numeric band of 4.0 on the tester's scoring scale. On assessment finding SEC-2026-10 the tester's evidence was that bucket configuration read from the cloud export, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. Low-severity finding SEC-2026-10 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Read alongside the remediation record, SEC-2025-12 against Analytics Warehouse Loader shows that the severity is right and the ownership is unclear, since two teams each believe the other holds it. The tester scored SEC-2025-12 against Analytics Warehouse Loader at 4.2 and labelled it Low. Analytics warehouse service account held write access to raw telemetry is the substance of Low-severity finding SEC-2025-12, raised on 2025-09-25. As at the review date SEC-2025-12 against Analytics Warehouse Loader was accepted and the closure entry reads not planned. finding SEC-2025-12 has no single accountable owner, and that is the finding the reviewer would press hardest.

7. Remediation of the licence exposure and its cost

What security supportability item TD-08 protects against is the fact that it no upstream security updates for a customer-facing component. The reviewer's qualification on TD-08 is that no advisory affects the current version, which is why it ranks low. The dependency structure places security supportability item TD-08 last in a three-item chain, so its date is the least reliable on the register. Remote Operations Console item TD-08 carries a follow-up request that was satisfied after the draft of TDR-2026-31 was circulated.

TD-01 covers the work to partition the fleet allocator write path. Scalability item TD-01 carries this observation: the load-test series identifies this as the single binding scalability constraint. Programme item TD-01 matters because it caps the region at roughly eight sites at current order density. TD-01 is reported here in full because a summary of it in an earlier draft of TDR-2026-31 was misread as an assurance.

The estimate for TD-17 on OTA Updater is 0.5 engineer-quarters and $72,000. Security item TD-17 carries this observation: the control is designed but the signing tool does not enforce it. OTA Updater carries register item TD-17, the work to enforce a two-person rule on OTA artefact signing. TD-17 would move from a concern to a verified defect if one further measurement were supplied.

2027-06-30 is the target for TD-03, whose status is backlog. Partly funded, with the second phase unbudgeted is the funding position on support item TD-03. The reviewer's comment on register item TD-03 is that management ranked this behind the separation monitoring work. Costmap Fusion item TD-03 is reported here in full because a summary of it in an earlier draft of TDR-2026-31 was misread as an assurance.

The register shows tooling item TD-07 as overdue, due 2026-05-31. Left in place, the condition behind TD-07 on Simulation Harness non-reproducible test results undermine every simulation claim. TD-07 on Simulation Harness carries this observation: the check is written but not enabled as at the review date. TD-07 appears twice in the working papers for TDR-2026-31 because the first answer was corrected after a follow-up request.

The reviewer's comment on TD-12 on Edge Gateway Agent is that the site alert landed first; the buffer extension needs a storage change on the compute module. Reading the programme plan, TD-12 is behind a supplier release outside the company's control. Resilience item TD-12 is on the register because it telemetry loss on outages beyond four hours. TD-12 on Edge Gateway Agent is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Programme item TD-16 is after the determinism work in the simulation harness, which is why the target date is later than the effort estimate implies. Security item TD-16 carries a 2026-10-15 target and a status of committed. For TD-16 on Localization Engine the team estimates 1 engineer-quarter, which the reviewer converted to $126,000 for comparison. register item TD-16 is the reason one section of TDR-2026-31 concludes narrowly rather than generally.

Whether TD-02 on Localization Engine happens depends on funding, and it is funded by reallocating effort from feature work. Register item TD-02 covers the work to replace the unmaintained scan-matching library in the localization fallback. The register shows IP and support item TD-02 as committed, due 2026-12-31. Localization Engine item TD-02 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Alarm Router carries programme item TD-15, the work to add per-region credentials for the paging provider. TD-15 on Alarm Router is sequenced last in a three-item chain, so its date is the least reliable on the register. Register item TD-15 appears in the register and is unfunded and dependent on capital rather than engineering effort. programme item TD-15 was checked against the corresponding entry in the company's own register, and the register was current.

Support item TD-09 is not straightforward: the upgrade requires a maintenance window at every site. Left in place, the condition behind register item TD-09 an unsupported broker underneath every service. The plan treats TD-09 on Event Bus as funded and already staffed, which the reviewer checked against the headcount model. TD-09 appears twice in the working papers for TDR-2026-31 because the first answer was corrected after a follow-up request.

Whether Localization Engine item TD-13 happens depends on funding, and it is funded and already staffed. The estimate for register item TD-13 is 1.5 engineer-quarters and $178,000. Carries the same dependency as the copyleft removal is the point the reviewer would put to management on Localization Engine item TD-13. Localization Engine item TD-13 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Bring the second bench rig to the current sensor configuration is entered at TD-05 on Hardware-in-the-Loop Rig as an test capability item. About register item TD-05 it should be said that outstanding since the April 2025 lidar timing incident. $96,000 is the costed estimate for register item TD-05, based on 0.5 engineer-quarters. programme item TD-05 rests on a single source, and the reviewer has said so rather than implying corroboration.

Raise robot software line coverage above sixty per cent on the perception path is entered at quality item TD-04 as an quality item. In progress is the status of register item TD-04, targeted at 2027-03-31. Sequencing of TD-04 on Costmap Fusion matters: it is behind a customer maintenance window that is not yet agreed. register item TD-04 is the reason one section of TDR-2026-31 concludes narrowly rather than generally.

The plan treats programme item TD-23 as funded from the second quarter onward, which the reviewer checked against the headcount model. The register entry TD-23 on Charge Scheduler reads: split the shared database instance used by the allocator and the charge scheduler, in the reliability category. If reliability item TD-23 is not done, the exposure is that it one noisy query affects task assignment. programme item TD-23 is included because a reader of TDR-2026-31 should be able to disagree with the inference without repeating the work.

8. Incidents involving a third-party component

Remediation for performance incident INC-2026-004 is to replace the fallback matcher and re-survey the affected aisles; Rahul Menon owns it and the committed date is 2026-06-30. Underneath Kestrel Medical Supply incident INC-2026-004 lies the design decision that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. The classification applied to Localization Engine incident INC-2026-004 is performance, and the affected component is Localization Engine. The detection path for performance incident INC-2026-004 was the site's commissioning engineer raised it during acceptance testing. Duration for performance event INC-2026-004 splits into not a service outage; 5 days to a workaround of impact and open at the review date to close the underlying cause. incident INC-2026-004 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

On performance event INC-2026-004 the evidence is strong on the cause and limited on the contributing factor; the severity assessment is stated separately below. 2026-01-13 is the date of Kestrel Medical Supply incident INC-2026-004, on which relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. On delivery, incident INC-2026-004 shows this: not landed; the matcher replacement is in design and the re-survey is scheduled. The affected population in performance event INC-2026-004 was one site and 18 robots. The contributing factor recorded against incident INC-2026-004 is that the site was surveyed with a sparse feature set that increases cache misses. INC-2026-004 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Process bypass incident INC-2025-131 was raised on 2025-10-14 after an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. The export restriction landed on 2025-11-27; the publication work is open is the position on the process bypass event INC-2025-131 commitment as at the review date. Classifying Juniper Bay Apparel incident INC-2025-131 as process bypass matters because the Map Authoring Studio backlog contains two further items of the same kind. Map Authoring Studio incident INC-2025-131 shows a short mitigation and a long tail: 3 h 22 m against 44 days. Scope of impact in Map Authoring Studio incident INC-2025-131 was one site and 34 robots, taken from the fleet records rather than from the narrative. Map Authoring Studio incident INC-2025-131 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The reviewer's reading of INC-2025-131 is that the incident is unremarkable in itself and revealing about the observability of the surrounding system. The corrective action for Juniper Bay Apparel incident INC-2025-131 is to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, and 2025-12-15 is the date the team gave. A contributing factor in INC-2025-131 was that the site had a standing workaround for slow map publication that predated the review step. Grading INC-2025-131 on evidence rather than on consequence gives moderate on the cause and strong on the delivery position, which was checked in the repository. Underneath process bypass event INC-2025-131 lies the design decision that the export function did not check publication state. process bypass event INC-2025-131 is the clearest instance in TDR-2026-31 of a control that exists on paper and is weaker in practice.

INC-2025-152 was caused by a design gap: the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. The detection path for Video Snippet Service incident INC-2025-152 was an internal privacy review found the discrepancy. Anjali Sekhon committed after Video Snippet Service incident INC-2025-152 to apply the lifecycle rule through the configuration baseline and delete the over-retained clips, due 2026-01-31. The immediate event in incident INC-2025-152 was that one region retained operator video clips for ninety days against a thirty-day commitment, dated 2025-12-04. The classification applied to data retention incident INC-2025-152 is data retention, and the affected component is Video Snippet Service. data retention incident INC-2025-152 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

In data retention incident INC-2025-152, bucket creation was not covered by the configuration baseline check, which is a process finding rather than a code finding. Internal felt internal incident INC-2025-152, which affected no production site and no robot directly, the exposure being to internal systems. As to whether the data retention event INC-2025-152 remediation actually landed: the rule landed on 2026-01-10; deletion of the historical clips is open. Internal incident INC-2025-152 rests on moderate on the cause and strong on the delivery position, which was checked in the repository evidence, which the reviewer states separately from the severity of the finding. Assessed on its own record, incident INC-2025-152 shows that the customer-reported detection path is the finding, not the defect itself. Video Snippet Service incident INC-2025-152 is treated as material to the investment case rather than as a routine engineering item.

The immediate event in incident INC-2025-058 was that two tenants reused the same order identifier namespace and duplicate suppression discarded live orders, dated 2025-05-09. INC-2025-058 reached the engineering team when the customer reported missing picks the same afternoon. The blast radius of INC-2025-058 was 2 site or tenant with no robot taken out of service. What went wrong in WMS Connector incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. Data integrity event INC-2025-058 belongs to the data integrity group and sits with WMS Connector. data integrity incident INC-2025-058 was one of two items where TDR-2026-31 could not obtain the underlying data within the engagement.

After data integrity event INC-2025-058, Tomas Villareal took the action to key suppression on the tenant and identifier pair and backfill the affected order records by 2025-07-15. Assessed on its own record, WMS Connector incident INC-2025-058 shows that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Data integrity event INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. The reviewer checked the incident INC-2025-058 commitment against the code and release records; landed on 2025-05-30; the backfill was completed on 2025-06-03. On incident INC-2025-058 the evidence is strong, resting on the production traces the team supplied; the severity assessment is stated separately below. Dunbar Industrial incident INC-2025-058 is treated as material to the investment case rather than as a routine engineering item.

Reported detection applies to scheduling event INC-2025-052, since the site's own shift report, escalated the next morning. The immediate event in INC-2025-052 was that an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet, dated 2025-04-21. After Evergreen Warehousing incident INC-2025-052, Nadia Semenova took the action to release charge windows on a shift-end event and reconcile the calendar nightly by 2025-06-30. Grading Charge Scheduler incident INC-2025-052 on evidence rather than on consequence gives strong on the cause and limited on the contributing factor. Scheduling event INC-2025-052 affected one site and 44 robots. scheduling incident INC-2025-052 is one of the entries where TDR-2026-31 relies on the company's own measurement rather than an independent one.

The reviewer checked the incident INC-2025-052 commitment against the code and release records; landed on 2025-06-11 with a nightly reconciliation report. The engineering team's root cause for Charge Scheduler incident INC-2025-052 is that the scheduler treated a shift calendar entry as immutable once the window had opened, and the reviewer found the analysis sound. Beyond the direct cause, scheduling incident INC-2025-052 shows that the site had changed its shift pattern without notifying the deployment team. Scheduling is the class of Evergreen Warehousing incident INC-2025-052; Charge Scheduler is the component. Evergreen Warehousing incident INC-2025-052 shows a short mitigation and a long tail: 4 h 10 m against 16 days. scheduling incident INC-2025-052 is included because a reader of TDR-2026-31 should be able to disagree with the inference without repeating the work.

The corrective action for data integrity incident INC-2025-110 is to fail the load on an incomplete partition and reprocess the affected days, and 2025-10-31 is the date the team gave. The post-mortem for data integrity incident

INC-2025-110 identifies the cause as follows: the loader treated a missing partition as empty rather than as incomplete. Scope of impact in incident INC-2025-110 was 1 site or tenant with no robot taken out of service, taken from the fleet records rather than from the narrative. The classification applied to INC-2025-110 is data integrity, and the affected component is Analytics Warehouse Loader. The reviewer's reading of data integrity incident INC-2025-110 is that the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. data integrity incident INC-2025-110 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The contributing factor recorded against data integrity incident INC-2025-110 is that the partition completeness check was written but never wired into the load gate. Detection came through the customer challenged the report three days later, which places INC-2025-110 on the reported side of the ledger. For data integrity incident INC-2025-110 the reviewer holds moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence and reports the finding at that strength. Duration for data integrity event INC-2025-110 splits into not applicable; reporting only of impact and 18 days to close the underlying cause. On 2025-09-03 the record for Kestrel Medical Supply incident INC-2025-110 opens: a late telemetry partition was excluded from the daily load and a customer report understated throughput. data integrity event INC-2025-110 is the clearest instance in TDR-2026-31 of engineering practice being better than the documentation suggests.

Telemetry Ingest accounts for multiple sites incident INC-2025-071, a availability event. The reviewer checked the availability event INC-2025-071 commitment against the code and release records; landed on 2025-07-18 and exercised in the September failover drill. Availability incident INC-2025-071 affected 6 sites and 214 robots. The judgement recorded against availability incident INC-2025-071 is that the incident is unremarkable in itself and revealing about the observability of the surrounding system. Erik Palmgren committed after incident INC-2025-071 to add a leadership and under-replicated partition check to the broker upgrade runbook, due 2025-07-31. incident INC-2025-071 is one of the entries where TDR-2026-31 relies on the company's own measurement rather than an independent one.

2025-06-11 is the date of incident INC-2025-071, on which a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. What went wrong in multiple sites incident INC-2025-071 was that a broker partition lost its leader during a rolling upgrade of the streaming cluster. The reviewer grades the INC-2025-071 evidence as moderate; the narrative is consistent but the reviewer could not independently reproduce it and has not upgraded the finding to compensate. 52 m elapsed before multiple sites incident INC-2025-071 was mitigated, with resolution at 52 m. Availability event INC-2025-071 would have been shorter or smaller had it not been that the upgrade runbook did not require a partition leadership check between broker restarts. Telemetry Ingest incident INC-2025-071 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

INC-2025-124 records that on 2025-10-01, a shard rebalance served shadow state up to ninety seconds stale without marking it stale. What incident INC-2025-124 tells the reviewer is that the customer-reported detection path is the finding, not the defect itself. Multiple sites felt Robot Shadow Store incident INC-2025-124, which affected 5 sites and 178 robots. Incident INC-2025-124 was detected because the console latency alert fired, though the underlying cause took a day to attribute. Consistency incident INC-2025-124 was mitigated in 1 h 08 m; the resolution ran to 29 days. Robot Shadow Store incident INC-2025-124 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Contributing to Robot Shadow Store incident INC-2025-124, and not repaired by the remediation, is that no test exercised a shard rebalance while the cluster was under load. For incident INC-2025-124 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. As to whether the incident INC-2025-124 remediation actually landed: landed on 2025-10-30; the load test was added on 2025-11-12. INC-2025-124 belongs to the consistency group and sits with Robot Shadow Store. Following multiple sites incident INC-2025-124 the team undertook to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman accountable by 2025-11-30. incident INC-2025-124 is the clearest instance in TDR-2026-31 of a control that exists on paper and is weaker in practice.

9. Performance dependence on third-party components

Measurement of run LT-2026-12 is against the 1,200 ms p95 objective published for order assignment rather than against a customer commitment. Scenario LT-2026-12 came out pass. 22 s at p95 is the figure that matters in scenario

LT-2026-12, against 1.4 s to console median and 48 s at the tail. load run LT-2026-12 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The environment behind scenario LT-2026-13 carries the current production topology with one region held out, and the reviewer weighted the result accordingly. The limiting term in load run LT-2026-13 was the allocator partitioning programme is the gating dependency. LT-2026-13, taken with the other runs in the series, shows that the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. twelve sites, projected 2027 fleet run LT-2026-13 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Drain took 71 minutes and delayed the analytics load past its window was the constraint run LT-2026-14 exposed. Four hour edge buffer drain after an outage is the scenario at scenario LT-2026-14: 720 robots, 13,800 orders an hour, dated 2026-04-15. In run LT-2026-14 the median was not applicable and the tail reached not applicable; errors ran at 0.00%. run LT-2026-14 is reported at the evidence strength available and has not been promoted to fit the narrative.

Configuration for 1000-robot run LT-2026-15 was 1000 robots against n/a orders an hour in the map publication to a thousand robots case. Comparing load run LT-2026-15 with its neighbour in the series, the difference is whether the read replicas were enabled. Measured latency in LT-2026-15 was 3 m 20 s at the median, 8 m 40 s at p95 and 13 m 10 s at p99. scenario LT-2026-15 was one of two items where TDR-2026-31 could not obtain the underlying data within the engagement.

Config distribution with thirty per cent of robots offline run LT-2026-16 exists in the series to isolate the presence of a failure injection. The recovery objective of forty seconds for a database failover is the target applied to run LT-2026-16. The constraint surfaced by config distribution with thirty per cent of robots offline run LT-2026-16 is offline robots applied on reconnect, which is the behaviour the February 2026 incident exposed. load run LT-2026-16 illustrates why TDR-2026-31 separates the severity of a finding from the confidence behind it.

Scenario LT-2025-11 is recorded as pass. The fifteen-minute map publication objective is the target applied to single site, steady state run LT-2025-11. The percentile set for 120-robot run LT-2025-11 reads 210 ms, 480 ms and 760 ms, with errors at 0.02%. single site, steady state run LT-2025-11 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

120-robot run LT-2025-12 is best read against the neighbouring run in the same series, from which it differs in the warehouse system endpoint used. LT-2025-12 was judged against the thirty-second telemetry visibility objective. Latency in LT-2025-12 was 265 ms at p50, 690 ms at p95 and 1,240 ms at p99, with an error rate of 0.05%. run LT-2025-12 is the entry that changed most between the first and second document requests behind TDR-2026-31.

A fleet simulator rather than physical robots, with the control loop unchanged describes the environment load run LT-2025-13 ran in. In single site, peak with a charger outage run LT-2025-13 the median was 310 ms and the tail reached 1,690 ms; errors ran at 0.11%. In scenario LT-2025-13 the bottleneck was charge scheduler retries added measurable queueing. load run LT-2025-13 is one of the entries where TDR-2026-31 relies on the company's own measurement rather than an independent one.

240-robot run LT-2025-14 returned 290 ms median, 820 ms at the 95th percentile and 1,510 ms at the 99th, erroring at 0.09%. The two sites, shared region test, recorded as load run LT-2025-14, was executed on 2025-10-14. Reading load run LT-2025-14 for the binding constraint gives no limit reached. 240-robot run LT-2025-14 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Measured latency in run LT-2025-15 was 355 ms at the median, 1,120 ms at p95 and 2,240 ms at p99. Four sites, shared region is the scenario at LT-2025-15: 480 robots, 9,400 orders an hour, dated 2025-10-16. Scenario LT-2025-15 is recorded as pass with observation. four sites, shared region run LT-2025-15 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

1,680 ms at p95 is the figure that matters in load run LT-2025-16, against 470 ms median and 3,410 ms at the tail. In six sites, shared region run LT-2025-16 the bottleneck was shadow store shard hotspotting on the two largest sites. Run LT-2025-16 came out fail against the 1,200 ms p95 target. six sites, shared region run LT-2025-16 is the clearest instance in TDR-2026-31 of a control that exists on paper and is weaker in practice.

Load run LT-2025-17 is not read as a headline number: the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. Before reading run LT-2025-17, note that the environment carries a fleet simulator rather than physical robots, with the control loop unchanged. Order broker journal write queue depth exceeded its alert threshold was the constraint 240-robot run LT-2025-17 exposed. run LT-2025-17 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Weighed against the projected fleet, 480-robot run LT-2025-18 indicates that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. Comparing 480-robot run LT-2025-18 with its neighbour in the series, the difference is the size of the concurrent operator session pool. The limiting term in load run LT-2025-18 was reproduced the November 2025 control delay before the topic split. LT-2025-18 was cross-checked against the engineering record supplied for TDR-2026-31, and the two agree.

Measurement of load run LT-2025-19 is against the 0.5 per cent error-rate ceiling in the internal service documentation rather than against a customer commitment. Against the stated targets, 480-robot run LT-2025-19 is fail against the error rate target. Scenario LT-2025-19 is the region failover during peak run, sized at 480 robots and 9,400 orders an hour. scenario LT-2025-19 is the clearest instance in TDR-2026-31 of engineering practice being better than the documentation suggests.

10. Licence questions left open at the review date

Three of the open items in TDR-2026-31 would close on a single document that the company has agreed to produce. Where the present report cannot conclude, it says what evidence would let it conclude. Selection for this report favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. TDR-2026-31 added two components to its sample after the first week, because the interview evidence pointed at them. S. Alderton notes that the company answered every document request, though two answers arrived after the draft.

S. Alderton asked in every declined case who would revisit the decision and when, and the present report records the answer even where there was none. Management's declinatures are reported in the report at TDR-2026-31 at face value; the reviewer's disagreement is stated separately and does not overwrite them. 15 follow-up items arose from the interview programme for TDR-2026-31; 2 were still open at delivery. Interviews for TDR-2026-31 were conducted with the owning engineer present rather than with a manager speaking on their behalf. The company reviewed a draft of TDR-2026-31 for factual accuracy and its corrections are reflected without altering any conclusion.

This report distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Report TDR-2026-31 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. Nothing in the present report supports a retention or an escrow; the findings are operational and they are funded. TDR-2026-31 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. S. Alderton has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Post-close monitoring under the report at TDR-2026-31 would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. S. Alderton would add one qualitative item to the pack proposed in the present report: the count of commitments closed without verifying evidence. This report sampled 17 services from an inventory of 34, weighted towards those on the critical control path. Sampling in the present report covered 17 of the 34 catalogued services and 6 of the 18 live sites. S. Alderton has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

11. Conclusions on intellectual-property hygiene

Pinehurst Systems Assurance concludes, on the material examined, that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. On the evidence available by 2026-07-01, the present report concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. For this report the reviewer sampled 6 sites, including the largest deployment and the newest. This report covers 17 of 34 services; the untested remainder is listed so a reader can see what was not reached. Report TDR-2026-31 contains 6 open items, and a reader should assume that number will move rather than settle.

The transaction reading of the present report is that the engineering risk is priced in the plan rather than in the purchase agreement. S. Alderton would put 7 items from report TDR-2026-31 into the first hundred days rather than into a warranty. The reviewer asked in each case whether a declined item had a compensating control, and the report at TDR-2026-31 reports the answer even where it is unsatisfactory. 2 declined items are carried in this report, of which two lack any documented compensating control. Each section of the report at TDR-2026-31 states its own basis, so a reader can accept one section and reject another.

4 indicators are recommended in the present report; the reviewer deliberately avoided proposing anything that needs a new pipeline. The pack report TDR-2026-31 proposes is deliberately short, because a longer one would be produced and not read. Coverage of report TDR-2026-31 is deliberately uneven: the safety and control path was examined closely and the reporting path only at a high level. 17 services were examined in detail for this report, chosen for their position in the dependency map rather than at random. Figures quoted in TDR-2026-31 come from the company's systems unless the text says the reviewer measured them.

Coverage for Remote Operations Console was 52 per cent line coverage at the review date, and the trend over the reviewed period was up by three points. Remote Operations Console was tested against the dependency ageing register, which records a charting library that no longer receives security updates. The published objective for Remote Operations Console is that operator input reaches the robot inside 350 ms at p95 on a supported network. Changes to Remote Operations Console reach production twice a week. The Remote Operations Console codebase is TypeScript with a React front end throughout, deployed onto cloud, a static bundle plus a session gateway, and its durable record is PostgreSQL 14 for session records and object storage for clips. Remote Operations Console was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Event Bus sits cloud, one cluster per region, three availability zones, and no alternative placement is currently supported. Event Bus is promoted upgraded twice a year, with the ring order fixed by policy. 4 defects remain open against Event Bus; the one the reviewer pressed on is that one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. The Event Bus codebase is operated open-source broker, no in-house code throughout, deployed onto cloud, a managed streaming cluster, and its durable record is the broker's own log storage with seven-day retention. Read against its objective, Event Bus the reviewer accepts the engineering position while noting that the verifying test does not exist. Event Bus is the clearest instance in TDR-2026-31 of a control that exists on paper and is weaker in practice.